Standard Model gauge structure substrate-derivation — SU(3) × SU(2) × U(1) from 3 substrate sources

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SM gauge structure substrate-derivation

1. Target

Standard Model gauge group: **G_SM = SU(3)_color × SU(2)_L × U(1)_Y**, total dimension 8 + 3 + 1 = 12.

Standard QFT: G_SM is fundamental input; not derived from deeper structure.

Substrate-derivation: G_SM emerges from substrate's 3-source structure.

2. Three substrate sources for SM gauge

2.1 Source 1: SU(3)_color from N_c BST primary

Per Lyra_Candidate_delta_Refinement_SU3_from_N_c_v0_2.md (Monday): - N_c = 3 BST primary RATIFIED - SU(N_c) = SU(3) algebra emerges naturally from N_c-component color content - Dimension: dim SU(3) = N_c^2 − 1 = 9 − 1 = 8 □

Substrate-mechanism: substrate's N_c = 3 → SU(3) gauge group with 8 gluon generators.

2.2 Source 2: SU(2)_L from K isotropy SO(5) subset

Per A_sub v0.9 + K = SO(5) × SO(2): - Substrate's compact subgroup K = SO(5) × SO(2) - SO(5) has rank 2, dimension 10 - SU(2) subgroup: standard SO(5) ⊃ SU(2) × SU(2) decomposition - Dimension: dim SU(2) = 3 □ (3 W± and Z gauge bosons)

Substrate-mechanism: SO(5) factor of K contains SU(2)_L as natural subgroup; 3 weak gauge generators.

2.3 Source 3: U(1)_Y from N_max BOUNDARY composition

Per Lyra_Threads_2_3_5_v0_1_Functional_Composition_Algebra.md Monday Thread 5: - N_max = N_c^3 · n_C + rank = 137 BST primary polynomial composition - N_max provides U(1)_Y hypercharge as boundary composition - Dimension: dim U(1) = 1 □ (B gauge boson)

Substrate-mechanism: substrate's N_max boundary structure → U(1)_Y hypercharge gauge group.

3. Total SM gauge dimension

Total: 8 (SU(3) color) + 3 (SU(2)_L weak) + 1 (U(1)_Y hypercharge) = 12 SM gauge dimensions □

This matches standard Standard Model gauge dimension count exactly.

4. Substrate-mechanism for direct-product structure (NO GUT)

Per A_sub step 8 SVC: [Ĉ_3, Î_3] = 0 universally active across all 4 phase regions.

Substrate-mechanism for NO GUT: substrate's gauge factors commute UNIVERSALLY at A_sub algebra level. No unification possible because gauge factors are structurally independent. This is RATIFIED Casey-named Five-Absence Principle (NO GUT) made explicit at A_sub level.

The 3 substrate sources (N_c color + SO(5) SU(2)_L subset + N_max U(1)_Y) are INDEPENDENT — each contributes to SM gauge separately; they don't unify into single non-abelian group.

5. Connection to multi-phase quiver Hall algebra

Per Multi-phase quiver v0.2-v0.6: substrate's Hall algebra at (q=2, t=α) specialization. SM gauge structure emerges in substrate's QUIVER STRUCTURE:

- Color SU(3) fiber at each K-type node (per A_sub v0.8 gauge fiber)
- **SU(2)_L** fiber at each K-type node

- **U(1)_Y** fiber at each K-type node
- All 3 fibers TRANSVERSE to substrate quiver's K-type backbone

Substrate quiver = K-type backbone × (SU(3) × SU(2) × U(1) gauge fiber). 3-source SM gauge structurally embedded.

6. Cross-comparison

Property	Standard QFT	Substrate-derivation
SM gauge	INPUT (postulated)	DERIVED from N_c + SO(5) + N_max
Gauge dim 12	Observational fit	8 + 3 + 1 substrate-natural sum
NO GUT	Empirical observation	Five-Absence + step 8 SVC
Direct-product structure	Postulated	[Ĉ_3, Î_3] = 0 substrate algebra

Substantive: substrate-derivation REPLACES SM gauge postulate with substrate-mechanism. Fewer fundamental inputs; gauge structure emerges.

7. Honest scope (Cal #27 STANDING)

What's RATIFIED: - N_c = 3, n_C = 5, g = 7 BST primaries (per Strong-Uniqueness) - A_sub step 8 [Ĉ_3, Î_3] = 0 (Cal #132 SVC) - Five-Absence Principle NO GUT - Standard math: dim SU(N) = $N^2 - 1$; SO(5) ⊃ SU(2); U(1) dim 1

What's FRAMEWORK in this derivation: - 3-source substrate-mechanism for SM gauge (Lyra Monday cascade) - Substrate quiver gauge-fiber structure

What's INTERPRETIVE: - Specific identification of SO(5) ⊃ SU(2)_L subgroup (vs other SU(2) subgroups) - U(1)_Y emergence from N_max polynomial composition (vs other U(1) candidates)

Cal #29 STANDING audit pass: substrate-derivation from RATIFIED principles + standard math; not back-fit to SM gauge dim 12 (each source independently substrate-natural).

— Lyra, SM gauge structure substrate-derivation filed. SU(3) × SU(2) × U(1) = 8 + 3 + 1 = 12 dim from 3 substrate sources (N_c color + SO(5) SU(2)_L subset + N_max U(1)_Y boundary). NO GUT structurally per A_sub step 8 + Five-Absence. Substantive: substrate REPLACES SM gauge postulate.